```
% ME 329 Lab 3 - Conveyor Gearmotor System Simulation
clear; clc;

fprintf('=== ME 329 Lab 3: Time-to-Speed Simulation with Load Inertia
===\n');
```

=== ME 329 Lab 3: Time-to-Speed Simulation with Load Inertia ===

=== System Parameters ===

```
v_belt_target = 470 * 12 / 60;    % Convert fpm to in/s
r_roller = 3;                     % Roller radius [in]
omega_roller = v_belt_target / r_roller;  % rad/s

fprintf('Target belt speed: %.2f in/s\n', v_belt_target);
fprintf('Required roller angular speed: %.2f rad/s\n', omega_roller);
```

Target belt speed: 94.00 in/s
Required roller angular speed: 31.33 rad/s

=== Gear Ratio and Motor Speed ===

```
G_total = 1/5.66;  % As provided by your professor
omega_motor_target = omega_roller / G_total;

fprintf('Selected gear ratio: %.2f (motor:roller)\n', G_total);
fprintf('This means the roller spins %.1fx faster than the motor.\n', 1 /
G_total);
fprintf('Required motor speed: %.2f rad/s\n', omega_motor_target);
```

Selected gear ratio: 0.18 (motor:roller)
This means the roller spins 5.7x faster than the motor.
Required motor speed: 177.35 rad/s

=== Motor Selection ===

```
motor_HP = 2;
rated_rpm = 1692;
T_full_load = 5252 * motor_HP / rated_rpm;

fprintf('Motor size: %.2f HP\n', motor_HP);
fprintf('Full-load torque: %.2f lbf·ft\n', T_full_load);
```

Motor size: 2.00 HP
Full-load torque: 6.21 lbf·ft

=== Motor Performance Curve (from spread-sheet) ===

```
pct_speed = [0, .20, .60, .70, .80, .85, .90, .94];
pct_torque = [3.00, 2.85, 2.40, 2.18, 1.85, 1.60, 1.30, 1.00];
```

```
rpm_vector = pct_speed * 1800;
omega_vector = rpm_vector * 2 * pi / 60;  % rad/s
torque_vector = pct_torque * T_full_load;
```

=== Conveyor Inertia and Friction ===

```
Jc = 40 / 12;              % lbf·in·s² → lbf·ft·s²
J_motor = 0.005;                % Motor inertia

% === Load Inertia Calculation (Point Mass at the Conveyor Radius) ===
mass_load = 325 / 32.2;         % Convert 325 lbf to slugs
r_conveyor = 0.25;              % 3 inches = 0.25 feet
J_load = mass_load * r_conveyor^2;

fprintf('Load inertia (moving with conveyor): %.5f lbf·ft·s^2\n', J_load);

% === Equivalent Inertia (Reflected to Motor Side) ===
Vc=470/60; %fpm to fps
Wfl = (1692 / 60) * 2 *pi; % full load rpm to rad/s
e = Vc/(Wfl*r_conveyor);
J_eq = J_motor + e^2*Jc + mass_load * (r_conveyor^2)*e^2;

F_fric = 0.3 * 325;             % Conveyor friction force
T_fric = F_fric * r_roller / 12;
T_fric_motor = T_fric * G_total;

fprintf('Reflected total inertia: %.5f lbf·ft·s^2\n', J_eq);
fprintf('Reflected friction torque: %.2f lbf·ft\n', T_fric_motor);
```

```
Load inertia (moving with conveyor): 0.63082 lbf·ft·s^2
Reflected total inertia: 0.12897 lbf·ft·s^2
Reflected friction torque: 4.31 lbf·ft
```

=== Compute Angular Acceleration ===

```
alpha = (torque_vector - T_fric_motor) ./ J_eq;
```

=== Trim to Valid α > 0 and Target Speed ===

```
valid_idx = omega_vector <= omega_motor_target & alpha > 0;
omega_valid = omega_vector(valid_idx);
alpha_valid = alpha(valid_idx);
```

=== Safe trapz() with check ===

```
if isempty(omega_valid) || numel(omega_valid) < 2
    warning('No region of positive acceleration exists under these
settings.');
    time_to_speed = NaN;
    max_speed_reached = 0;
```

```
    percent_speed = 0;
else
    % === Integration Logic ===
    % Time to reach target speed
    time_to_speed = trapz(omega_valid, 1 ./ alpha_valid);
    max_speed_reached = omega_valid(end);
    percent_speed = max_speed_reached / omega_motor_target * 100;

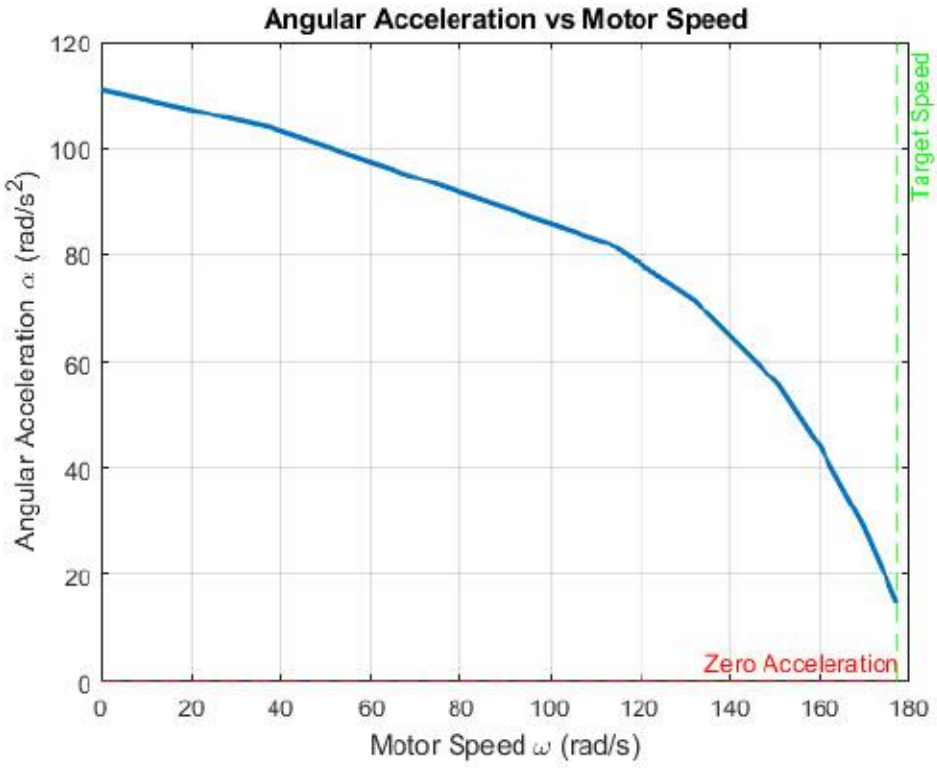
    fprintf('\n--- RESULTS ---\n');
    fprintf('Max motor speed reached with positive α: %.2f rad/s\n',
max_speed_reached);
    fprintf('Percent of target motor speed achieved: %.1f%%\n',
percent_speed);
    fprintf('Estimated time-to-speed (partial): %.3f seconds\n',
time_to_speed);

    if percent_speed < 100
        warning('Motor could not reach full target speed. Consider
increasing HP or adjusting G.');
    else
        fprintf('Target speed fully achieved within %.3f seconds.\n',
time_to_speed);
    end
end
```

```
--- RESULTS ---
Max motor speed reached with positive α: 177.19 rad/s
Percent of target motor speed achieved: 99.9%
Estimated time-to-speed (partial): 2.568 seconds
Warning: Motor could not reach full target speed. Consider increasing HP or
adjusting G.
```

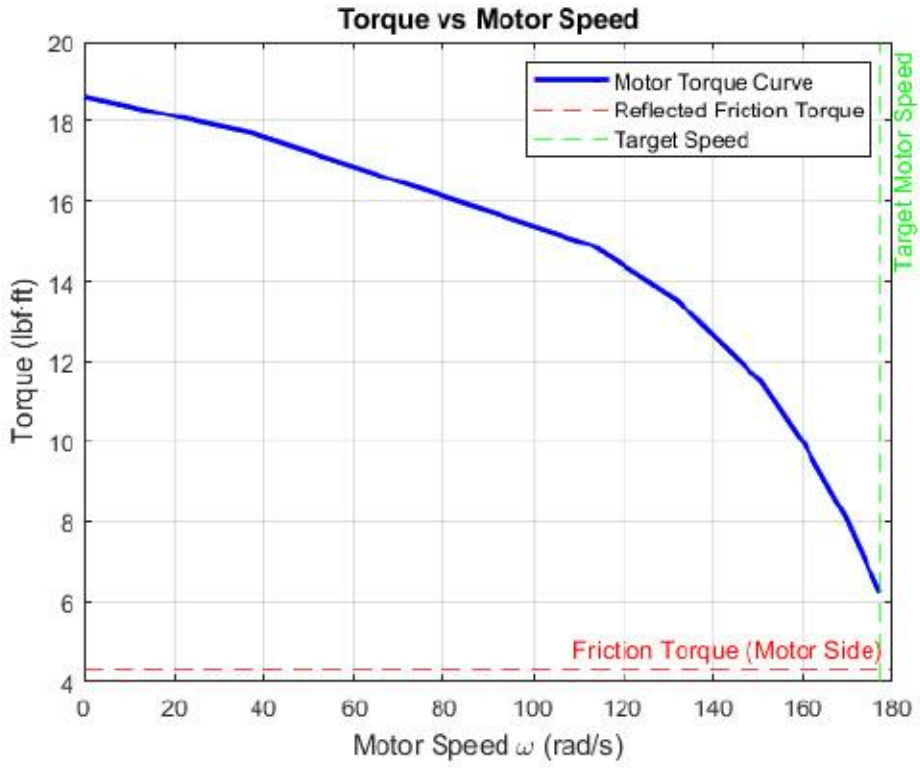
=== Plot Acceleration Curve ===

```
figure;
plot(omega_vector, alpha, 'LineWidth', 2); hold on;
yline(0, 'r--', 'Zero Acceleration');
xline(omega_motor_target, 'g--', 'Target Speed');
xlabel('Motor Speed \omega (rad/s)');
ylabel('Angular Acceleration \alpha (rad/s^2)');
title('Angular Acceleration vs Motor Speed');
grid on;
```



=== Plot Torque-Speed Curve ===

```
figure;
plot(omega_vector, torque_vector, 'b-', 'LineWidth', 2); hold on;
yline(T_fric_motor, 'r--', 'Friction Torque (Motor Side)');
xline(omega_motor_target, 'g--', 'Target Motor Speed');
xlabel('Motor Speed \omega (rad/s)');
ylabel('Torque (lbf·ft)');
title('Torque vs Motor Speed');
legend('Motor Torque Curve', 'Reflected Friction Torque', 'Target Speed');
grid on;
```



Published with MATLAB® R2024b